INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

AIR CONDITIONER A/E32C-24

This specification is approved for use within Code 84, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a 60/50-Hz, electric-motor-driven, military, compact, vertical, lightweight air conditioner, designated A/E32C-24, having a nominal cooling capacity of 36,000 BTU/hr and a nominal heating capacity of 28,600 BTU/hr.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood.

Military

MIL-P-116	Preservation, Methods of.
MIL-A-38339	Air Conditioner, Lightweight, Compact, Military, General Requirements for.
MIL-D-38386	Duct Assembly, Ground, Conditioned Air, Insulated, Flexible.
MIL-C-38392	Compressor, Hermetic, Air Conditioning, General Specification for.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4120

AMSC N/A

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2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

DRAWINGS

Goodyear Aerospace Corp.	Shelter Assembly
3142110-001-109 and details	S-530 A/G Shelter
Trane Company	
DL A4524-5450-4	Model A/E32C-24 Air Conditioner, Trane Model MAC40 Reprocurement Data Package.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification , however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The requirements of MIL-A-38339 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.1.1 Drawings. The requirements of Drawing list DL A4524-5450-4 and detail drawings thereto apply as requirements of this specification with the exceptions and additions specified herein (see 6.3).

3.2 Design and construction

3.2.1 Electric power source. The air conditioner shall be designed to operate from an alternating-current power source having the following characteristics:

a.	Potential (steady)	208V +10 -5 percent
b.	Frequency	60 H_Z ±5 percent or 50 Hz ±3 percent
c.	Phase	3 phase, 4 wire.

3.3 Performance

3.3.1 Cooling capacity. The preproduction units shall have a minimum cooling capacity of 38,000 BTU/hr and the production units shall have a minimum of 36,000BTU/hr when operating on the 60-Hz frequency. At any rated capacity, the total power draw shall not exceed 10.3 kw.

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3.3.2 Heater capacity. The power consumption of the electric resistance heater shall be 8,400w ±5 percent, exclusive of the fan motor.

3.4 Evaporator assembly. The evaporator assembly shall be physically interchangeable (in the air conditioner) with the evaporator assembly depicted on the drawings (see 3.1.1) and shall meet the performance requirements specified herein. If required in order to meet the dehumidification requirements and tests of MIL-A-38339, a mist eliminator shall be included as part of the evaporator assembly.

3.4.1 Evaporator airflow. When the air conditioner is operating under the 60 Hz frequency current and in 125°F ambient and with 90°F dry-bulb and 75°F wet-bulb return air to the evaporator, the evaporator fan shall provide a minimum of 1,290 cfm at free-air delivery and a minimum of 980 cfm against an external static pressure of 1 inch of water.

3.5 Condenser assembly. The condenser assembly shall be physically interchangeable (in the air conditioner) with the condenser assembly depicted on the drawings (see 3.1.1) and shall meet the performance requirements specified herein.

3.6 Compressor. The compressor shall meet the performance requirements of MIL-C-38392 as specified for class 4, type I compressors.

3.7 External flexible ducts. When specified (see 6.2), two 12-inch-diameter external flexible ducts in accordance with MIL-D-38386 shall be provided with each air conditioner.

3.7.1 Duct adapter. When ducts are required, an adapter shall be furnished to provide a transition between the air conditioner evaporator supply and return air openings and the ducts. The adapter shall conform to the drawing specified in MIL-A-38339.

3.8 Canvas cover. When specified (see 6.2), an external canvas cover shall be furnished in accordance with the drawings (see 3.1.1).

3.9 Noise level. The noise level generated by the air conditioner shall not exceed the values specified in table I when tested in accordance with MIL-A-38339 and as specified herein.

3.9.1 Shelter. The air conditioner shall be mounted in an S-530 A/G , 60 Hertz shelter in accordance with Goodyear Aerospace Drawing 314-2110-001-109 and subdrawings.

3.10 Dimensions. The dimensions of the air conditioner shall be as follows:

	Inches
Width	30 ± 1/16
Depth	18 ± 1/16
Height	55 ± 3/32

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*TABLE I

Maximum Noise Level Limits

Octave Band Center Frequencies (Hz)	Sound Pressure Level in Decibels (Ref. 0.0002 Dynes/CM^2)	
	Evaporator Side	Condenser Side
63	86	81
125	86	81
250	85	81
500	82	81
1,000	78	77
2,000	75	73
4,000	67	66
8,000	57	58

3.11 Weight. The weight of the complete air conditioner charged with refrigerant and oil shall not exceed 445 pounds.

3.12 Workmanship. The air conditioner shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions; thoroughness of soldering; et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection and testing. The inspection and testing of the air conditioner shall be in accordance with MIL-A-38339 and as specified herein.

4.3 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY.

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5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The air conditioner shall be preserved and packaged in accordance with method IIA of MIL-P-116.

5.1.2 Level C. Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The air conditioner preserved and packaged as specified in 5.2.2 shall be packed in a style B, type II, class 2 container conforming to PPP-B-601. Skids shall be provided to facilitate machine handling. As far as practical, containers shall be of uniform size and shape and of minimum tare and cube consistent with the protection required. Closure and strapping shall be in accordance with the appendix of the specification.

5.2.2 Level B. Level B packing shall be the same as level A, except the container shall be style B, type III, class 2 of PPP-B-601. Skids shall be provided to facilitate machine handling.

5.2.3 Level C. The air conditioner preserved and packaged as specified in 5.1.2 shall be packed in a manner to ensure carrier acceptance and safe delivery at destination. Containers shall be in accordance with either the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. The air conditioner shall be cushioned, anchored, blocked, and braced in accordance with MIL-STD-1186. The Freefall drop test, Edgewise drop tests, and Cornerwise drop tests shall be in accordance with appendix A of MIL-STD-1186. The tests are not required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Air Conditioner A/E32C-24

6. NOTES.

6.1 Intended use. The A/E32C-24 air conditioner is intended for use in cooling, dehumidifying, heating, filtering, and circulating air for portable shelters or vans in accordance with electronic and personnel requirements. The air conditioner may be installed indoors or directly exposed to the atmosphere.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When external flexible ducts are to be provided (see 3.7) and required lengths.

c. When a canvas cover is required (see 3.8).

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

e. Special shipment marking (see 5.4).

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6.3 Drawings. Included in Trane Company DL A4524-5450-4 are the latest known revisions of all drawings available as applicable to the items specified. The drawings are not warranted by the Government to meet the specified performance requirements. In addition, the drawings are not warranted as being a full set or sufficiently detailed for complete manufacturing purposes. The contractor is advised to carefully evaluate all drawings furnished for bidding purposes prior to preparation of his proposal or bid. It is the contractor's responsibility to evaluate this data and prepare his quotation on the basis of a comprehensive understanding of the task to be performed. The specification represents the minimum needs of the Government. The drawings furnished represent efforts of the Government to define the requirements of the specification. Wherever appropriate, revisions or changes to the drawings shall be made to meet the performance requirements of the specification. However, changes to the drawings that affect the interchangeability of parts, assemblies, or subassemblies specified on the drawings will not be made without prior approval from the procuring activity. In the event, and to the extent approved, it becomes necessary to change drawings to meet the performance requirements of the specification, such costs of changes will be borne by the contractor and no equitable adjustment will be authorized. The inability of the Government to provide additional data or the incompleteness of any data furnished hereunder shall not later be construed as justification for an increase in contract price or as an excusable delay under the terms of the contract clause entitled Default.

6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

BTU/hr .....British thermal units per hour

cfm .....cubic feet per minute

Hz.....Hertz, cycles per second

kw.....kilowatt

V .....volt

w.....watt

6.5 Subject Term (Key Word) Listing.

Air Conditioner
Heater, Electric Resistance
Shelter, S-530 A/G Shelter

Custodian
Air Force - 99

Preparing activity
Air Force - 84

Project No: 4120-F318

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